1. Some things to think about in the afternoons and beyond

(1) Understand the details of the consequences of subconvexity mentioned in lecture (reference: [19, §5]):
   - Subconvexity vs. geometry of numbers.
   - Distinguishing modular forms.
   - Duke theorem. Reduction of supersingular elliptic curves.
   - Quantum unique ergodicity. One exercise here is to understand how this follows from subconvexity in the special case of Eisenstein series, as in [17, §2].

(2) Some "classic" papers (non-exhaustive):
   - Bounds for Fourier coefficients of half-integral weight, applications to quadratic forms: [15], [9], [11]
   - Subconvexity for GL_2: [10] [12] [13].
   - Moments and amplification via periods: [27], [20], [16], [24].
   - Shifted convolution sums:
     - via δ-symbol: [10]
     - via periods: [25], [4], [5].
   - Papers emphasizing variation of the test vector: [23, 2, 27].

   One exercise is draw parallels, e.g., between
   - [20, Thm 5.1] and [10],
   - [27, §4] and [12], or
   - [20, Thm 5.2] and [13].

   Another is to reprove some results using different methods, e.g., by working out a "classical" proof in the style of [13] for subconvexity for Maass forms at special points, namely, for $L(1/2+it_f, f)$ with f on $\mathrm{SL}_2(\mathbb{Z})$ of eigenvalue $1/4+t_f^2$, by estimating an amplified fourth moment, e.g.,

$$\sum_{f: t_f \in [T, T+1]} \left| \sum_{\ell \asymp L} c_\ell \lambda_f(\ell) \right|^2 \left| L(\tfrac{1}{2} + it_f, f) \right|^4.$$

(3) Study the proof of the convexity bound. There are two steps:
   - The Phragmen–Lindelöf convexity principle, to reduce estimates for $\Re(s) = 1/2$ to estiamtes for $\Re(s) = 1+\varepsilon$ and $\Re(s) = -\varepsilon$.
   - The functional equation, to reduce further to estimates for $\Re(s) = 1+\varepsilon$.
   - Establishing the necessary bounds for $\Re(s) = 1+\varepsilon$, for which see `https://www.math.wsu.edu/faculty/scliu/papers/Convexity.pdf` and references.

(4) Some recent papers, concerning subconvexity or related problems, that haven't been fully explored (e.g., interpreted via integral representations):
   - δ-method papers such as [26] and [1]
   - Higher moments over very large families, as in [7], [8]
   - Rankin–Selberg when the rank difference is larger than one, as in [6]

(5) Higher rank subconvex bounds [3], [18], [22], [21], [14]. There are many "exercises" implicit in these papers; for instance, a half-dozen are suggested in [22, Remark 1.4]. Some other questions:
   - These have all proceeded via arithmetic amplification. Is it possible to succeed in some cases via "family shortening" (as in, e.g., [25])? A natural case to try would be the t-aspect. Some experiments with GL_2

suggest this is difficult (see `https://ultronozm.github.io/math/20230522T174726__shrinking-archimedean-families-second-moment-gl2.html`).

"Purely horizontal" aspects remain open, e.g., twists by Dirichlet characters of prime conductor on GL_4.

(6) (Extra credit) Create a song of thematic relevance to the lectures. Examples:
   - `https://suno.com/playlist/a5ddede1-8601-4cba-acb3-dea7abdc1b81`
   - `https://suno.com/song/4b267a84-2268-4fe7-a499-42e56b34526b`, `https://suno.com/song/001a15e5-29d0-4fab-98c2-3aaf2257fe92`

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